

OLA Booking Data Analysis

1. Project Overview

This project analyzes OLA ride-booking data for May 2024 (100,000 bookings). The goal is to uncover insights into booking patterns, revenue performance, ride cancellations, vehicle type preferences, and customer and driver ratings to guide operational and business decisions.

The analysis was done in three stages: Python (pandas) for data exploration, PostgreSQL for structured business queries, and a 5-page interactive Power BI dashboard for visual reporting.

2. Dataset Summary

- Rows: 100,000
- Columns: 16
- Time Period: May 2024 (01-05-2024 to 31-05-2024)
- Unique Customers: 60,288 | Unique Booking IDs: 99,456
- Vehicle Types: 7 (Auto, Mini, Prime Sedan, Prime SUV, Prime Plus, Bike, E-Bike)
- Locations: 50 unique pickup and 50 unique drop locations
- Booking Statuses: 5 (Success, Cancelled by Customer, Cancelled by Driver, Other, Incomplete Ride)
- Key Features:
 - Booking details (Date, Time, Booking ID, Booking Status, Customer ID, Vehicle Type)
 - Trip details (Pickup Location, Drop Location, Ride Distance, Booking Value in INR)
 - Service and quality metrics (Avg VTAT, Avg CTAT, Driver Rating, Customer Rating)
 - Cancellation and incomplete-ride flags with reasons (by customer / by driver)
- Missing Data: Avg VTAT and Avg CTAT are blank for rides that were not completed successfully

3. Exploratory Data Analysis using Python

We began with data loading and exploration in Python:

- **Data Loading:** Imported Booking_ola.csv using pandas (pd.read_csv).
- **Initial Exploration:** Used df.head(10) to preview records and df.describe() for summary statistics.
- **Dataset Shape:** Confirmed 100,000 rows and 16 columns.
- **Unique Values:** Checked categorical fields: 5 booking statuses, 7 vehicle types, 50 pickup and 50 drop locations.

- **Missing Data Handling:** Avg VTAT and Avg CTAT contain nulls only for non-successful rides. These are expected, since no turnaround time exists when a ride is not completed.
- **Data Quality Check:** Booking ID has 99,456 unique values against 100,000 rows, so about 544 Booking IDs repeat. Worth reviewing for duplicates before final reporting.
- **Database Integration:** Loaded the cleaned data into PostgreSQL for SQL analysis.

Metric	Min	Max	Mean
Avg VTAT (mins)	1	15	~7.99
Avg CTAT (mins)	1	12	~6.50

- **Most frequent values:** Booking Status = Success (62,094 rides); Vehicle Type = Auto (25,077 bookings); Pickup = Pisoli; Drop = Wanowrie.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Successful Bookings** – Counted rides completed successfully.
Result: 62,094 rides (62.09% of all bookings).
2. **Average Ride Distance by Vehicle Type** – Compared average trip distance across the 7 vehicle types.

Vehicle Type	Avg Distance (km)
Bike	18.43
Prime SUV	18.41
Prime Plus	18.34
Mini	18.31
Prime Sedan	18.21
E-Bike	18.12
Auto	18.09

Distance is almost identical (~18 km) for every vehicle type.

Rides Cancelled by Customers – Counted bookings cancelled by the customer.

Result: ~6,872 rides (6.87% of bookings).

Top 10 Customers by Ride Count – Ranked customers by number of bookings in May 2024.

Result: The top customer made 7 bookings in the month.

3. **Driver Cancellations by Reason** – Broke down driver-side cancellations by reason.

Driver Cancellation Reason	Share of Driver Cancellations
The customer was coughing / sick	25.21%
Customer related issue	25.12%
More than permitted people in the vehicle	24.91%
Personal & Car related issues	24.76%

The four reasons are almost evenly split, so no single cause dominates.

4. **Max and Min Driver Rating (Prime Sedan)** – Checked the rating range for Prime Sedan drivers.

Result: Max 5.0, Min 1.0 (re-check the minimum against your query output; ratings in the dashboard average ~4.0).

5. **Busiest Booking Hour** – Found the hour of day with the most bookings using the time column.

Result: 8 AM – 9 AM (Peak bookings)

Average Customer Rating by Vehicle Type – Compared customer ratings across vehicle types.

Result: All vehicle types sit at ~4.0 (range 3.99 to 4.01).

6. **Total Booking Value of Successful Rides** – Summed the fare amount for completed rides.

Result: INR 32.42M from successful rides.

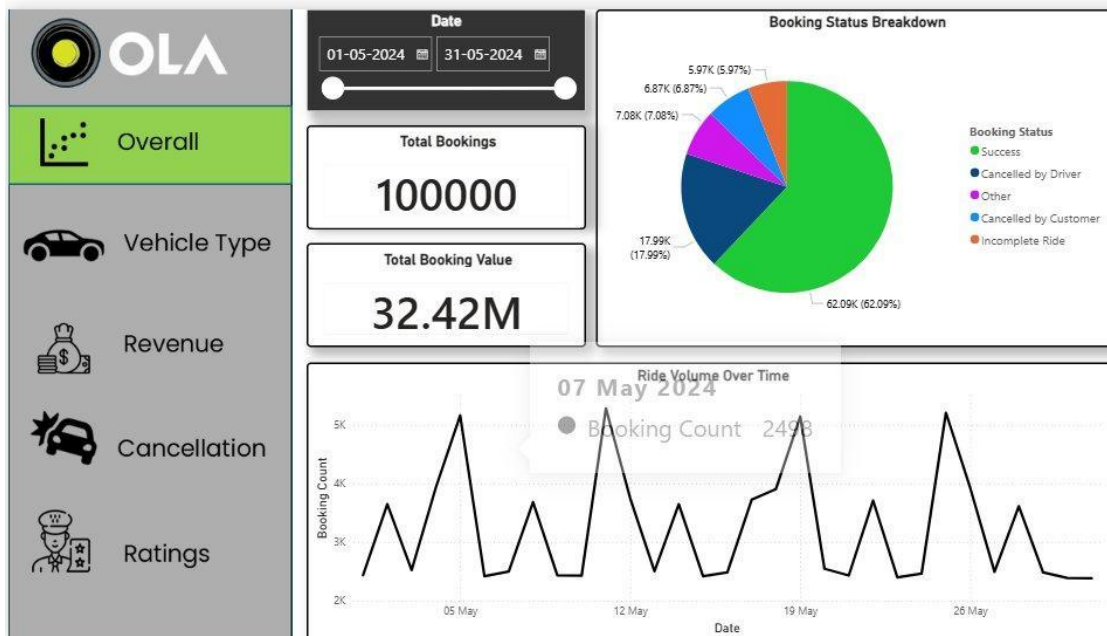
7. **Incomplete Rides with Reasons** – Listed incomplete rides along with the recorded reason.

Result: 5.97% of bookings were incomplete. Driver not moving to pickup (23.34% of incomplete rides)

5. Dashboard in Power BI

Finally, we built an interactive 5-page dashboard in Power BI with a date slicer that filters every page.

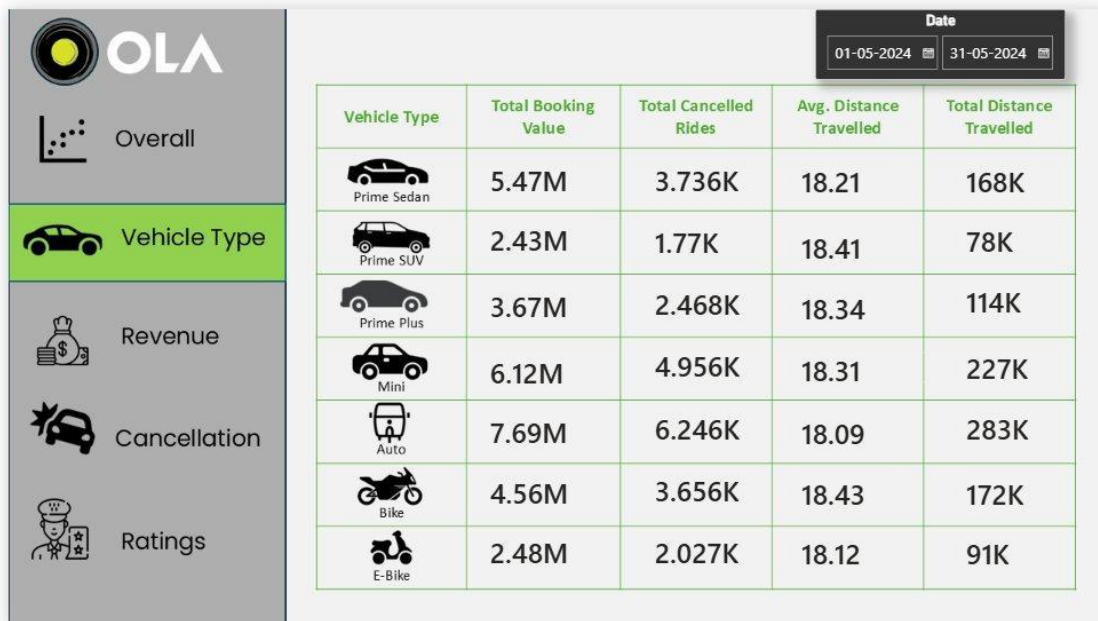
Page 1: Overall



Dashboard: Overall page

- Total Bookings: 100,000. Total Booking Value: INR 32.42M (booking value is recorded for successful rides).
- Status split: Success 62.09%, Cancelled by Driver 17.99%, Other 7.08%, Cancelled by Customer 6.87%, Incomplete 5.97%.
- Daily bookings swing between roughly 2.4K and 5K, with a recurring peak about once a week.

Page 2: Vehicle Type



Dashboard: Vehicle Type page

Vehicle Type	Booking Value (INR)	Cancelled Rides	Avg Distance	Total Distance
Auto	7.69M	6,246	18.09 km	283K
Mini	6.12M	4,956	18.31 km	227K
Prime Sedan	5.47M	3,736	18.21 km	168K
Bike	4.56M	3,656	18.43 km	172K
Prime Plus	3.67M	2,468	18.34 km	114K
E-Bike	2.48M	2,027	18.12 km	91K
Prime SUV	2.43M	1,770	18.41 km	78K

- Auto is the top vehicle type by bookings (25,077) and booking value (INR 7.69M), followed by Mini (INR 6.12M).
- Prime SUV and E-Bike generate the least value (INR 2.43M and 2.48M).
- Auto also has the most cancellations (6,246), but that is about 24.9% of its bookings, in line with the platform average of 24.86%. The high count reflects volume, not a worse cancellation problem.

Page 3: Revenue



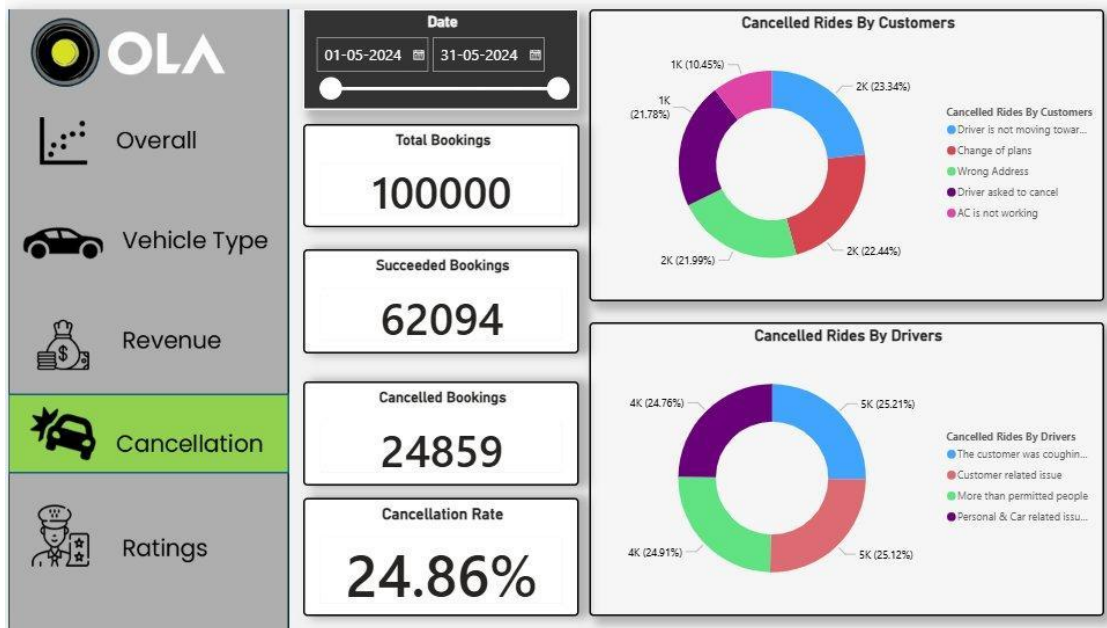
Dashboard: Revenue page

Rank	Customer ID	Booking Value (INR)
1	CUST20370	6,573
2	CUST71195	6,045
3	CUST90261	5,803
4	CUST57433	5,699
5	CUST35180	5,250
	Total (as shown on dashboard)	29,371

Individual values add to 29,370; the dashboard total of 29,371 reflects rounding of decimals.

- Highest revenue: Auto (INR 7.69M), then Mini (INR 6.12M) and Prime Sedan (INR 5.47M).
- Revenue by Day peaks at about INR 2M on days 5, 11, 19 and 25 of May, against a typical INR 0.7M to 1.3M on other days.
- The top 5 customers account for under 0.1% of total booking value, so revenue is spread across a very wide customer base.

Page 4: Cancellation



Dashboard: Cancellation page

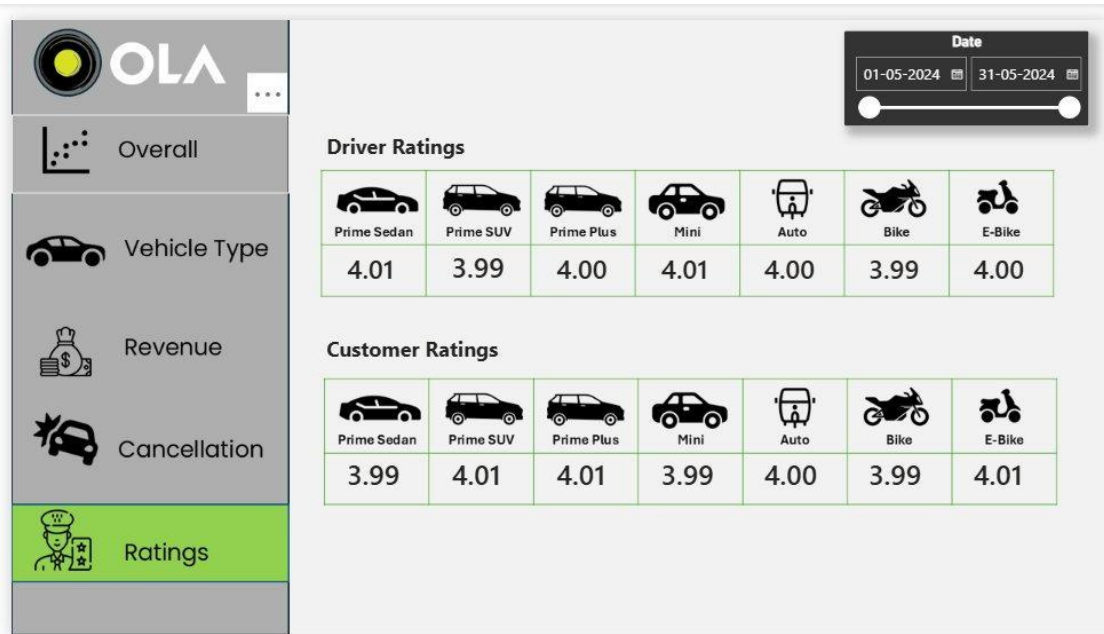
- Total Bookings: 100,000 | Succeeded: 62,094 | Cancelled: 24,859 | Cancellation Rate: 24.86% (driver + customer cancellations divided by total bookings).
- Driver-side cancellations (17.99%) are about 72% of all cancellations.

Cancelled by Customers: Reason	Share
Driver is not moving towards pickup	23.34%
Change of plans	22.44%
Wrong address	21.99%
Driver asked to cancel	21.78%
AC is not working	10.45%

Cancelled by Drivers: Reason	Share
The customer was coughing / sick	25.21%
Customer related issue	25.12%
More than permitted people	24.91%
Personal & Car related issues	24.76%

- Customer cancellations: driver not moving towards pickup leads at 23.34%, but the top four reasons are close. Together, "driver not moving" and "driver asked to cancel" make up about 45%.
- Driver cancellations are split almost evenly across four reasons (24.8% to 25.2% each).

Page 5: Ratings



Dashboard: Ratings page

Vehicle Type	Driver Rating	Customer Rating
Prime Sedan	4.01	3.99
Prime SUV	3.99	4.01
Prime Plus	4.00	4.01
Mini	4.01	3.99
Auto	4.00	4.00
Bike	3.99	3.99
E-Bike	4.00	4.01

- Driver and customer ratings are between 3.99 and 4.01 for every vehicle type.
- Ratings are stable at about 4.0 out of 5, but the gap between vehicle types is tiny, so ratings do not separate good and poor segments.

6. Business Recommendations

- **Reduce Driver Cancellations** – Drivers cause about 72% of cancellations, spread evenly over four reasons, so one fix will not be enough. Combine cancellation penalties and acceptance incentives, pre-shift vehicle checks, and clearer rules and in-app flows for rider count and rider-related issues.
- **Fix the Pickup Experience** – About 45% of customer cancellations involve the driver (not moving to pickup, or asking to cancel). Improve GPS guidance in the driver app and flag drivers who accept a ride but do not move.
- **Reduce Wrong-Address Cancellations** – Wrong address is 21.99% of customer cancellations. Add pickup pin confirmation in the booking flow.
- **Focus on Auto and Mini Fleets** – They bring in the most revenue. Increase availability on the weekly peak days (around 5, 11, 19 and 25 May).
- **Peak Hour Deployment** – Use the busiest-hour analysis to place more drivers in high-demand periods and cut wait times.
- **Broad-Based Loyalty Programs** – Top 5 customers are under 0.1% of revenue, so reward repeat riders in general rather than a few individuals.
- **Grow Prime SUV and E-Bike Demand** – They have the lowest booking values. Use targeted promotions to lift bookings.
- **Investigate Incomplete Rides** – 5.97% of bookings end incomplete, which is lost revenue. Find and fix the root causes.